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Cloud data privacy protection with homomorphic algorithm: a systematic literature review

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Abstract

This article critically examines the increasing reliance on cloud computing for data storage and processing, with a focus on the associated security and privacy challenges, especially when handling sensitive data. It explores homomorphic encryption (HE), particularly fully homomorphic encryption (FHE), as a promising solution for enabling computations on encrypted data without compromising confidentiality. Through a systematic literature review of studies published between 2017 and 2024, the article evaluates various homomorphic encryption algorithms, assessing their effectiveness in encryption and decryption, as well as performance metrics such as execution time, communication overhead, and energy consumption. The findings identify FHE as a widely researched technique, yet highlight significant challenges, including high computational overhead and complexity, which limit its suitability for real-time applications. The article underscores the need for continued research to enhance the efficiency of HE algorithms and calls for collaborative efforts among academia, industry, and regulators to establish robust frameworks for data privacy and security in cloud computing environments.

Keywords Cloud, Data, Privacy protection, Homomorphic, Security, Systematic Literature Review

Introduction

The massive generation and utilization of data on a daily basis present both challenges and opportunities [57]. Preserving this vast volume of data necessitates modern approaches to storage, security, integrity, privacy, and availability. Cloud computing has emerged as a transformative solution, addressing many of these concerns. However, it also introduces notable challenges such as security and privacy risks, interoperability and portability limitations, reliability and availability concerns, resource management and scheduling difficulties, energy

consumption, virtualization complexity, and performance and bandwidth costs [104]. Among these, security encompassing confidentiality, integrity, and availability remains a primary concern for both cloud service providers and consumers. Data security, in particular, is a critical issue in cloud computing environments, requiring continuous innovation and robust solutions to protect sensitive information from unauthorized access and breaches.

According to the National Institute of Standards and Technology (NIST), cloud computing is defined as a model that enables convenient, on-demand network access to a shared pool of configurable computing resources—such as networks, servers, storage, applications, and services—that can be rapidly provisioned and released with minimal management effort or service provider interaction [11].

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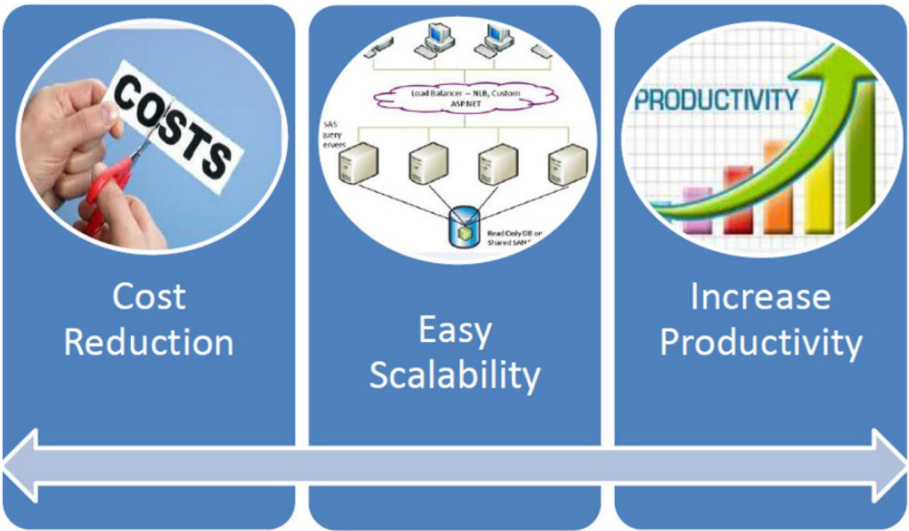


Fig. 1 Key benefits of cloud adoption

In pursuit of greater efficiency, governments world-wide are rapidly adopting cloud computing [4]. Similarly, businesses increasingly view cloud adoption as a strategic IT solution to reduce costs and leverage scalable data processing capabilities. Figure 1 illustrates the key benefits of cloud adoption. Cloud computing is a transformative technology that delivers computing resources as a service rather than a product on demand via the Internet.

The widespread adoption of cloud computing by businesses and governments has led to the emergence of major cloud service providers, including Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM, Huawei, Alibaba, Tencent, and Watson [93]. As of 2024, AWS, Google Cloud, and Microsoft Azure remain the top three providers globally.

The rapid integration of cloud computing into commercial operations has made cloud environment security a critical concern. In this context, where data is often distributed across multiple locations and managed by third-party providers, safeguarding sensitive information from unauthorized access and breaches presents significant challenges. Regulatory compliance—such as the EU’s General Data Protection Regulation (GDPR), the U.S. Health Insurance Portability and Accountability Act (HIPAA), and the Payment Card Industry Data Security Standard (PCI-DSS) is also essential, as violations can result in severe financial penalties and reputational damage [55]. Additionally, limited visibility and transparency in cloud operations hinder effective control, making it difficult for organizations to monitor and manage cloud-based resources securely.

Cloud computing models		
Service models	Deployment models	Essential characteristics
1. Software as a Service (SaaS)	1. Private cloud	1. On-demand self-service
2. Platform as a Service (PaaS)	2. Community cloud	2. Broad network access
3. Infrastructure as a Service (IaaS)	3. Public cloud	3. Resource pooling
	4. Hybrid cloud	4. Rapid elasticity
		5. Measured service

Cloud computing models

According to Hussain et al. [45], Cloud Computing Model consists of five characteristics, three service models, and four deployment models as summarized in Table 1.

Service models

SaaS (Software as a Service) Software as a Service (SaaS) is a cloud-based software distribution model in which applications are hosted by a vendor or service provider and accessed by users over the Internet. SaaS eliminates the need for end users to install and maintain software locally. Common examples include email services, Amazon Chime, calendaring tools, and Microsoft Office 365 [93]. Figure 2 demonstrate some use cases for SaaS.

PaaS (Platform as a service) Platform as a Service (PaaS) enables the development, deployment, and management of applications by providing a comprehensive computing platform along with essential programming tools and infrastructure. PaaS allows users to build and run applications without managing the underlying hard-

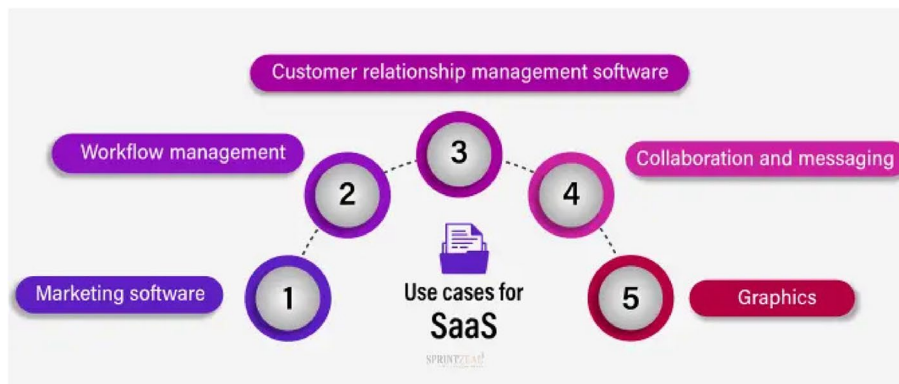


Fig. 2 Use cases for SaaS

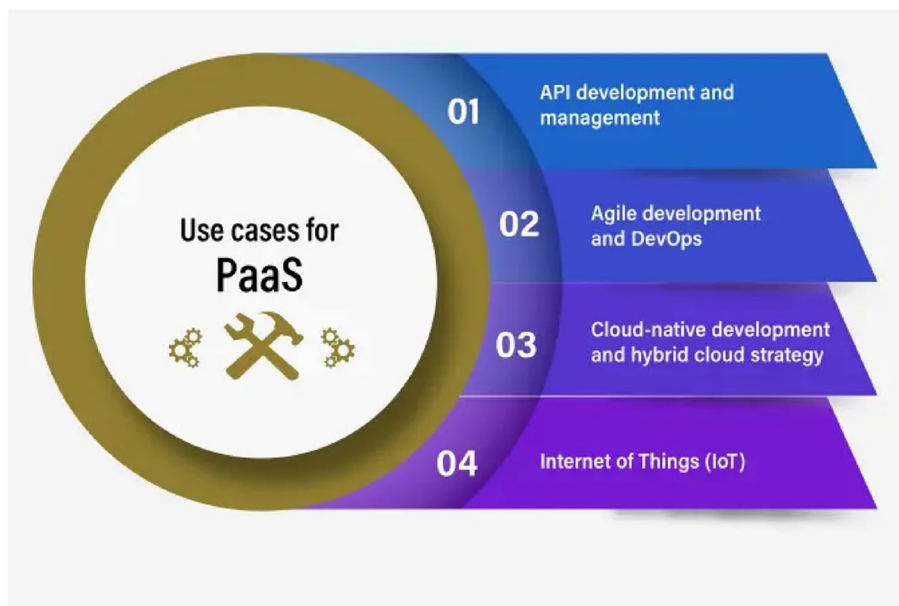


Fig. 3 Use cases for PaaS

ware and software layers [75]. By leveraging development tools, operating systems, and infrastructure provided by cloud service providers, users can focus on application functionality rather than system maintenance. Figure 3 illustrates key use cases for PaaS. Examples include Big-Query, Azure App Service, Azure Functions, Azure Logic Apps, AWS Elastic Beanstalk, and AWS App Service—all hosted in the cloud and accessed via the Internet.

IaaS (Infrastructure as a service) Infrastructure as a Service (IaaS) is a fundamental cloud computing model that provides on-demand access to core computing resources such as storage, servers, networking components, and virtualization layers. It eliminates the need for users to purchase, configure, or manage physical infrastructure on-premises. IaaS is ideal for temporary, experimental, or dynamic workloads, offering cost-effective, scalable, and flexible deployment options. Each user installs their

own operating systems and applications while leveraging virtualized resources over the Internet. Services typically include IP addresses, load balancers, bandwidth, and virtual server space, all sourced from distributed data centers [56].

Examples of IaaS: Microsoft Azure, Amazon Web Services (AWS), Cisco Metacloud, Google Compute Engine (GCE) (Fig. 4).

Container as a Service Container as a Service (CaaS) is a cloud computing model positioned between Infrastructure as a Service (IaaS) and Platform as a Service (PaaS) in terms of control and abstraction. Its lightweight architecture, compared to traditional virtual machines, has accelerated its adoption in recent years. CaaS provides a platform for deploying, managing, and orchestrating containerized applications without the need to configure the underlying infrastructure. It offers developers and IT

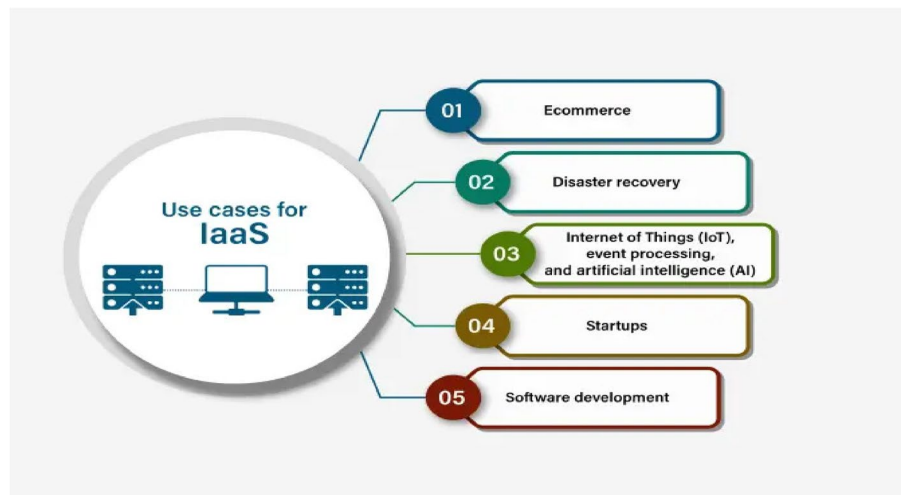


Fig. 4 Use cases for IaaS

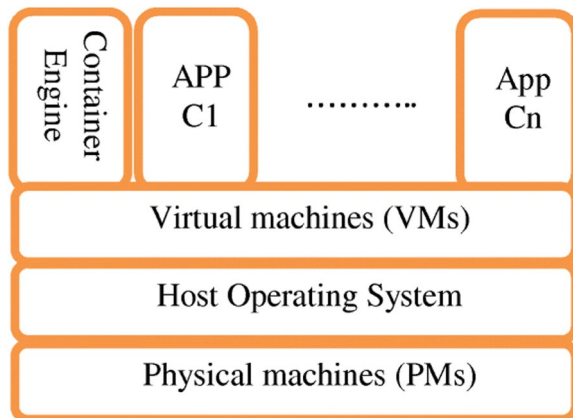


Fig. 5 Container as a service architecture [46]

teams enhanced flexibility and efficiency, with built-in tools such as Kubernetes and Docker Swarm for orchestration, as well as features like automated scaling, load balancing, and failover [89, 90]. Figure 5 illustrates the architecture of a typical CaaS environment.

Some of the popular Container as a service provider are:

- Amazon Elastic Kubernetes Service (EKS)—AWS's Kubernetes-based service.
- Google Kubernetes Engine (GKE)—Google's managed Kubernetes offering.
- Azure Kubernetes Service (AKS)—Microsoft's container management service.
- Red Hat OpenShift—An enterprise Kubernetes platform.
- Docker Hub—Simplifies container management and deployment.

Deployment models

Cloud deployment models define how cloud services are delivered, managed, and accessed. The main models public, private, community, and hybrid offer varying levels of control, security, and scalability, each suited to specific organizational requirements and regulatory needs.

- **Public Cloud:** Public cloud services are delivered over the internet to multiple clients, offering scalability and cost-efficiency, though often raising concerns about data security and control. The infrastructure is fully owned and managed by the cloud service provider [76]. These environments are affordable, easy to deploy, and ideal for testing and development scenarios. Leading providers such as AWS, Azure, and Google Cloud offer virtual machines, storage, and managed database services.
- **Private Cloud:** Dedicated infrastructure for a single organization, enhancing security and control but often at a higher cost [65]. Deployed using OpenStack to create a private cloud environment with full control over the infrastructure and configurations.
- **Community Cloud:** A community cloud is a shared infrastructure model used by multiple organizations with common goals or compliance requirements. It promotes collaboration while offering moderate control over data and security. Although often marketed as a distinct model, the underlying infrastructure may technically be a private, hybrid, or virtual private cloud. This model balances resource sharing with tailored governance [76].
- **Hybrid Cloud:** Combines public and private clouds, allowing for flexibility and scalability while addressing specific security need [76]. Combined both public and private cloud resources to simulate

Table 2 Performers in the cloud and the functions [18]

Performer	Functionality
Cloud Service Consumer	This entity utilizes cloud services, such as applications and storage, to meet their IT needs
Cloud Service Provider	Responsible for delivering cloud services, this performer manages the infrastructure and applications that consumers access
Cloud Broker	Acts as an intermediary, facilitating the selection and integration of cloud services from multiple providers to optimize consumer choices
Cloud Auditor	Ensures compliance and security by evaluating the services and operations of cloud providers, thus maintaining trust
Cloud Carrier	Provides the connectivity and transport of data between consumers and providers, ensuring reliable service delivery

a hybrid environment, leveraging the flexibility of public cloud and the control of private cloud.

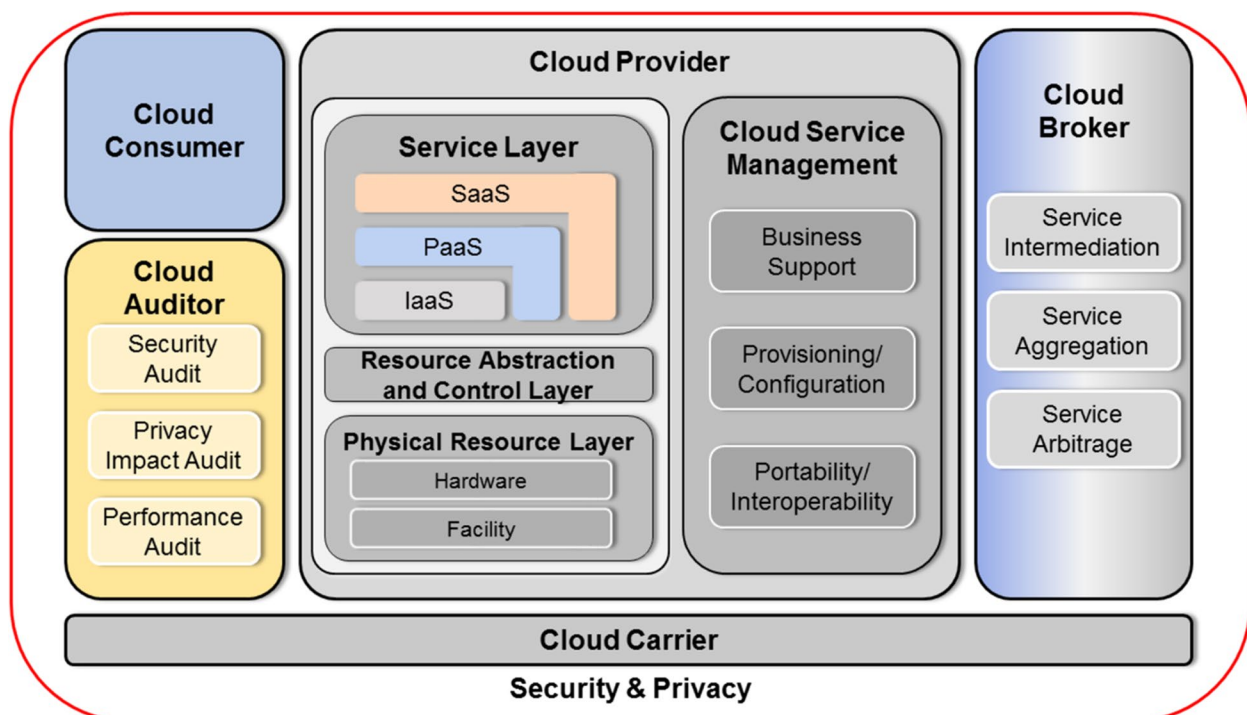
Understanding cloud deployment models enables organizations to choose the best fit for their operational and security needs. However, evolving cloud technologies bring new security challenges that must be proactively managed [74]. Key measures—such as encryption, access control, and Security Information and Event Management (SIEM) are implemented to enhance security. These environments are tested through simulated cyberattacks to evaluate the effectiveness and resilience of each control.

Cloud client A cloud client is a device or application that connects to cloud services, allowing users to access cloud resources using devices like PCs, smartphones, or embedded systems [57].

Cloud service providers A cloud service provider (CSP) delivers computing resources and services over the internet, enabling remote data access without physical infrastructure. CSPs offer flexible and cost-effective solutions to organizations of all sizes, typically on a pay-as-you-go basis. Their services include software, storage, and processing power. CSPs operate using service models such as Software as a Service (SaaS), Infrastructure as a Service (IaaS), and Platform as a Service (PaaS). They support remote collaboration and productivity by allowing access from any location with an internet connection [17]. Additionally, CSPs ensure data security, compliance, and provide complete infrastructure planning and management. A detailed functionality of performers in the cloud is depicted in Table 2 according to the Cloud Computing Reference Architecture proposed by NIST in Fig. 6.

The National Institute of Standards and Technology (NIST) identifies five key performers in cloud computing, each playing a distinct role within the ecosystem. These performers are essential for understanding the interactions and functionalities of cloud services.

Cloud computing resource management is essential for effective service delivery and optimized performance. Key components include resource allocation strategies,

**Fig. 6** Cloud Computing Reference Architecture proposed by NIST

such as dynamic allocation to meet real-time demand [6], virtualization to improve resource utilization [80], and energy-efficient models like Decentralized Energy-Aware Collaborative Model (DEACM) for minimizing consumption [80]. Scheduling techniques enhance performance, with task scheduling and load balancing preventing bottlenecks and improving consistency [6, 67]. Challenges involve scalability and resource contention, which complicate management as services expand [62, 67]. Figure 6 shows resources at the Service provider level and resource at the user level.

Security challenges in cloud services Cloud computing introduces a new model for delivering a wide range of computer services to many users anytime and anywhere. While it offers several benefits, it also brings significant risks. The primary concerns involve privacy and security issues such as unauthorized access, data leaks, and exposure of sensitive information [45]. Threats like viruses, trojan horses, and both internal and external hackers can compromise cloud infrastructure and launch widespread attacks. Classifying these threats allows businesses to prioritize security measures and allocate resources effectively. Major risk areas in cloud computing include network security, user security, software vulnerabilities, and especially data security.

Security in Platform as a Service (PaaS)

PaaS security is becoming more critical due to evolving cybersecurity threats. Studies have identified key vulnerabilities affecting PaaS environments. API weaknesses remain a major concern due to poor user awareness and lack of training [79]. Loss of access control when relying on CSPs introduces trust and compliance challenges [48]. Additionally, the continuous evolution of cyber threats demands strong and adaptive security frameworks [29].

Innovative solutions have been proposed to strengthen PaaS security. Fuzzy-Based Access Control applies fuzzy logic to boost performance and enhance protection [77]. Signature-Based Authentication uses public and private key signatures to secure data and maintain integrity. This method has shown a marked improvement in key security metrics ("Security Enhancement and PaaS Platform Authentication in Cloud Computing Based on SBA", 2022).

Security in Infrastructure as a Service (IaaS)

IaaS security involves threat identification, access control, encryption, and monitoring to maintain data confidentiality and compliance [69]. Credential management, authentication, and session handling are essential to prevent unauthorized access [50]. Vulnerability assessment is a critical aspect of securing IaaS platforms. The expansion of web applications has increased exposure to risks like SQL injection and XSS attacks. Implementing customized VAPT frameworks helps detect and mitigate

these vulnerabilities [49]. A holistic security strategy incorporating encryption, access control, and monitoring ensures resilience against evolving threats and supports business continuity [71].

Security in Software as a Service (SaaS)

SaaS security is a major concern due to its multi-tenant architecture and shared cloud resources. Key vulnerabilities include data privacy, weak authentication, and risks during data processing and storage, which require strong encryption and compliance [87]. The interconnected nature of SaaS systems adds complexity and increases the risk of data breaches and integration issues [88]. Compared to on-premises solutions, SaaS may face higher risks due to differences in how vendors prioritize security spending [44]. Despite its benefits, SaaS demands continuous development of advanced security measures to mitigate its inherent vulnerabilities.

Task scheduling in a hybrid cloud environment involves optimizing task allocation across diverse resources, as shown in Fig. 7. This optimization improves performance, reduces costs, and ensures efficient resource use. Various advanced algorithms and models have been developed to tackle the complexities of hybrid clouds. The hybrid fractional flamingo beetle optimization (FFBO) method, combining deep learning and multi-objective optimization, enhances makespan, energy consumption, and resource utilization [53]. Dynamic scheduling models using Continuous-Time Markov Decision Process (CTMDP) enable real-time adjustments to maximize cloud provider rewards [73]. Additionally, hybrid meta-heuristic techniques, such as merging the Moth Swarm Algorithm and Chameleon Swarm Algorithm, optimize task distribution while maintaining security through advanced encryption [10].

Cloud security issues Since large amounts of data are transmitted over networks and stored in cloud services, several vulnerabilities can be exploited by malicious actors [96]. Major cloud security concerns focus on data privacy, integrity, and compliance, requiring robust measures to protect sensitive information. The list below shows some key challenges related to cloud security issues

Data privacy and confidentiality

Data privacy and confidentiality are critical concerns in cloud computing as businesses increasingly store and process Sensitive information in the cloud. Advances like cryptography and confidential computing have significantly strengthened data security. Confidential computing, through TEEs, reduces data breaches by 50% and improves consumer trust by 40% in financial institutions [85]. It isolates sensitive data, preventing access even by cloud administrators and reducing the attack surface [84]. Cryptographic methods enhance cloud security while aiming to improve runtime and memory

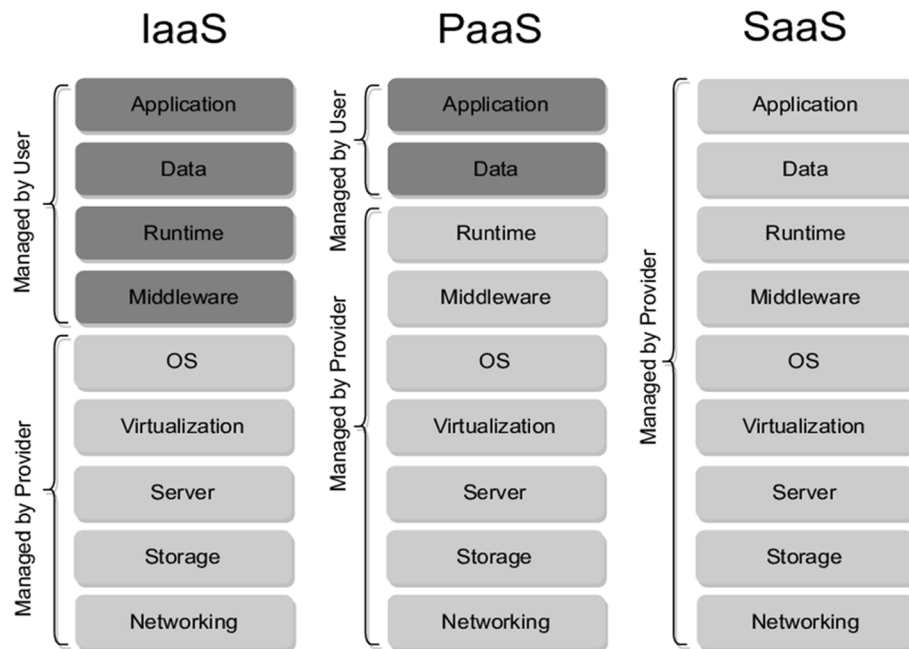


Fig. 7 Cloud computing resource management

performance [25]. A classification-based algorithm dynamically encrypts data based on Sensitivity, improving processing speed by 10% [95]. ID-PUIC allows clients to verify data integrity remotely without downloading full datasets [78], though some performance and implementation challenges still exist.

Identity and Access Management (IAM)

Identity and Access Management (IAM) is fundamental in cloud computing, enabling secure access based on user attributes [9]. Cloud IAM systems face challenges due to interoperability, scalability, and complex user environments, necessitating advanced authentication to prevent insider threats and unauthorized access [94]. Standard IAM protocols like SAML, OAuth, and OpenID Connect play a key role in secure access control [70]. Mobile Cloud Computing (MCC) requires specialized evaluation criteria to assess IAM performance under resource constraints. Security as a Service enhances IAM scalability and standardization [36], while dual-factor authentication ensures mutual verification between users and servers [47]. The PRIAM scheme and decentralized IAM models offer privacy-preserving and scalable solutions [103]. Alsirhani, Ezz and Mostafa, [9] demonstrated the effectiveness of IAM methodologies in protecting cloud platforms against a wide range of security threats.

Vulnerabilities in cloud infrastructure

Cloud-based IIoT systems are vulnerable to serious threats such as VM escape, hyperjacking, and side-channel attacks [12]. VM escape allows co-location attacks in shared infrastructures, posing a significant data security risk [32]. Even with Virtual Private Clouds, co-location

threats are increasing in frequency and success [2, 102]. Tools like Cloud Radar and the Online Penetration Suite help identify risks, while hypervisors remain prime targets due to their control over multiple VMs [59]. Although weak IaaS security can escalate into broader cloud vulnerabilities [41], proper security management can still ensure robust protection.

Homomorphic encryption (HE) enhances privacy and security in Medical IoT (MIoT) systems by enabling computations on encrypted data stored in the cloud. Recent works integrate HE with blockchain for secure data sharing [3] and use it in privacy-preserving federated learning [100]. Paillier-based HE schemes also support secure medical image encryption for telemedicine [107]. Additionally, the HealthLock system leverages HE and blockchain to protect sensitive health data [42]. These advancements reduce privacy risks while maintaining data utility in MIoT applications.

Lack of standardization

The lack of standardized security frameworks makes it difficult for enterprises to evaluate cloud security, forcing them to assess risks independently [23]. This complicates the implementation of effective SLAs and highlights the inadequacy of traditional corporate security policies in cloud environments [21]. Open standard identity protocols such as SAML, OAuth, and OIDC are essential for secure, cross-organizational identity verification without compromising sensitive data (Naik and Jenkins, 2017). While the absence of uniform standards remains a challenge, current efforts aim to develop consistent frameworks and protocols to strengthen cloud security.

Proposed techniques used to secure the cloud Numerous cutting-edge methods have been put forth to protect cloud systems, with an emphasis on improving user authentication, intrusion detection, and data integrity. These techniques make use of sophisticated frameworks and algorithms to handle the escalating security issues in cloud computing.

Intrusion Detection Systems (IDS)

Intrusion Detection Systems (IDS) are critical in cloud computing for defending Sensitive data against evolving threats, with recent improvements driven by machine learning and deep learning methods. Techniques like RBFNN for intrusion detection and RF for feature selection achieved over 92% accuracy and significantly boosted MCC scores [19]. SSAFS-DLID and lightweight CNN models with PSO further improved detection performance, reaching accuracies of 99.71% and F1-scores of 91.90%, respectively [83, 86]. Additional models like BERT classifiers and ensemble methods including Random Forest and boosting algorithms have consistently achieved over 99% accuracy across multiple datasets [8, 98].

Cryptographic techniques

Cryptographic methods play a crucial role in ensuring data integrity and preventing unauthorized access in cloud environments. They enhance the authenticity

and confidentiality of sensitive data stored in the cloud through advanced encryption techniques. Symmetric algorithms like AES and asymmetric algorithms like the McEliece cryptosystem ensure secure data access [22, 92]. The PRKCHB approach improves data security metrics by up to 9%, offering better integrity and confidentiality than traditional methods [91]. Steganography enhances protection by hiding encrypted data in images [39]. Techniques like fuzzy keyword search and fingerprint identification secure access to encrypted files [16]. Key management and evolving threats remain major challenges to cryptographic security. Block ciphers encrypt data in fixed-size blocks and resist known attacks [28]. Stream ciphers encrypt data bit-by-bit and are optimized for real-time applications [13].

Homomorphic encryption (HE) is a cryptographic technique that allows computations to be performed directly on encrypted data without needing to decrypt it first. The result of such computations remains encrypted and, when decrypted, matches the outcome of operations performed on the plaintext. This property enables secure data processing in untrusted environments, such as cloud computing, by ensuring data confidentiality throughout computation (Table 3).

Homomorphic encryption schemes are categorized based on the types of operations they support:

- Partially Homomorphic Encryption (PHE): Supports only one type of operation, such as addition or multiplication.
- Somewhat Homomorphic Encryption (SHE): Allows a limited number of operations before the noise in the ciphertext becomes too great.
- Fully Homomorphic Encryption (FHE): Supports unlimited operations on ciphertexts, enabling complex computations.

1. Paillier Cryptosystem
Supports Additive Homomorphism, use to Secure voting, private data aggregation.

$$E(m_1) \cdot E(m_2) \bmod n^2 = E(m_1 + m_2)$$

2. RSA Cryptosystem (in raw form)

- Supports: Multiplicative Homomorphism
- Operation:

$$E(m_1) \cdot E(m_2) \bmod n^2 = E(m_1 \cdot m_2)$$

3. Boneh–Goh–Nissim (BGN)

Table 3 Summary of homomorphic encryption schemes

Category	Example Schemes	Supported Operations	Security Weakness
PHE	Paillier, RSA, ElGamal, Goldwasser–Micali	One type only (Addition or Multiplication)	Efficient but limited in functionality which increases security risk, vulnerable to Chosen-Ciphertext Attacks (CCA)
SHE	BGN, Gentry [35]	Limited Additions and Multiplications	Improper selection of noise bounds or modulus sizes may weaken security or allow partial plaintext recovery. Since full bootstrapping is not used, attackers could infer limits on supported computations and exploit it for information leakage
FHE	Gentry [35], BFV, BGV, CKKS, TFHE	Unlimited Additions and Multiplications	Complex Parameter Space can lead to weak configurations, Most FHE schemes depend on the hardness of LWE/RLWE problems, which are currently secure but could be weakened by future quantum computing or algorithmic advancements

Operations Supported:

- You can add encrypted values as many times as needed.
- You can do only one multiplication between two encrypted values.

Example:

- You encrypt two numbers: E(3) and E(5)
- You can do:
 - $E(3) \times E(5) \rightarrow E(3 + 5) = E(8)$ (addition)
 - Then: $E(8) \times E(2) \rightarrow E(8 + 2) = E(10)$
 - But: $E(3) \cdot E(5) \rightarrow E(3 \times 5) = E(15)$ (multiplication once)
 - You cannot multiply again after that

4. BFV Homomorphic Operations (Simplified Notation)

$m_1 \cdot m_2$ = plaintext messages (integers modulo t)

$Enc(m)$ = encryption of message m

$Dec(c)$ = decryption of ciphertext c

+ and $\times$ represent homomorphic addition and multiplication on ciphertexts

1. Homomorphic Addition

$$Enc(m_1) + Enc(m_2) = Enc(m_1 + m_2 \mod t)$$

Adding ciphertexts corresponds to addition of plaintexts modulo t .

After decryption:

$$Dec(Enc(m_1) + Enc(m_2)) = m_1 + m_2 \mod t$$

2. Homomorphic multiplication

$$Enc(m_1) \times Enc(m_2) = Enc(m_1 \times m_2 \mod t)$$

Multiplying ciphertexts corresponds to multiplication of plaintexts modulo t .

After decryption:

$$Dec(Enc(m_1) \times Enc(m_2)) = m_1 \times m_2 \mod t$$

3. Noise growth and limits

Each operation increases noise inside the ciphertext. BFV can perform many additions and multiplications before noise becomes too large. When noise is too high, bootstrapping or ciphertext refresh is needed to continue (Table 4).

Noise management in HE with bootstrapping

Noise management is vital in homomorphic encryption (HE) to maintain computation accuracy, and bootstrapping is key to reducing accumulated noise in ciphertexts. Guimarães and Pereira [38] introduced a fast amortized bootstrapping method that handles multiple messages at once, lowering both noise and key size while enhancing performance through sparse polynomial multiplication. Another 2025 study on the FINAL FHE scheme used ellipsoidal Gaussian sampling and larger decomposition bases to cut bootstrapping noise by 76% and memory use by 47%. These methods significantly improve the practicality of HE systems. Overall, such innovations enhance the scalability and efficiency of secure encrypted computations.

User Authentication

Cloud security depends heavily on strong user authentication to protect sensitive data. Multi-factor authentication (MFA) improves security by requiring multiple identity verification steps. This layered method reduces the risk of unauthorized access and cyber threats. Static IP address techniques further enhance authentication and access control precision [16].

Multi-Factor Authentication (MFA) Implementation

MFA integrates various authentication factors, such as biometrics, tokens, and passwords, to create a more secure environment [15]. The implementation of MFA has shown to improve threat mitigation and reduce liability for data breaches, enhancing overall user experience [15].

Role of Blockchain in Authentication

Integrating blockchain with MFA enhances cloud security by ensuring immutable credential verification and increasing trust [27]. This approach has proven effective in detecting unauthorized access with higher precision and accuracy [27]. Comprehensive cloud protection strategies also involve secure coding, regular updates, and machine learning-based threat detection [56]. However, usability remains a challenge, as complex systems may negatively impact user experience.

Identified Problem

Cloud computing offers scalable virtual storage and processing but introduces significant security risks. Control of organizational data by CSPs creates vulnerabilities

Table 4 Identify the underlying hardness assumptions HE

Scheme	Operations Supported	Hard Problem	Security Basis	Key Size (bits) According to NIST 2025 report	Noise Growth Rate	Noise Control Techniques
Paillier	Addition	Composite Residuosity	IND-CPA	2048 (Acceptable (112-bit security)), 3072 (Strong (128-bit security)), 4096 + (High (Above 128-bit))	Homomorphic operations in Paillier do not increase “noise” in a way that threatens decryption [34]	Bootstrapping, Strategies like modulus switching and operator depth control can be adopted but not a necessary requirement
RSA	Multiplication	Integer Factorization	Only secure with padding	2048 (Acceptable (112-bit security)), 3072 (Strong (128-bit security)), 4096 + (High (Above 128-bit))	Does not experience noise growth during operations	RSA HE is a multiplicative partial HE scheme, its inherent noise is minimal and largely predictable, so no dedicated noise-control techniques
BFV/BGV	Addition and Multiplication	Ring-LWE	IND-CPA	1. 128-bit modulus q around 2^{34} – 2^{40} lattice dimension $n \approx 2^{12}$ (≈ 4096) 2. 192-bit Larger q and n (e.g. dimension $n \approx 8192$ – 16384) 3. 256-bit Even larger (e.g. $n \geq 16,384$ – $32,768$)	Noise increases with each homomorphic operation, particularly multiplication (quadratic effect over depth)	Modulus switching: reduces noise by scaling ciphertext to a smaller modulus after operations Relinearization/key-switching: helps control noise during multiplications; more advanced analysis factors in dependencies correctly [26]
CKKS	Addition and Multiplication (approximate arithmetic)	Ring-LWE (approximate)	Approximate IND-CPA	1. 128-bit modulus q around 2^{34} – 2^{40} , lattice dimension $n \approx 212$ (≈ 4096) 2. 192-bit Larger q and n (e.g. dimension $n \approx 8192$ – $16,384$) 3. 256-bit Even larger (e.g. $n \geq 16,384$ – $32,768$)	Each homomorphic operation adds noise: Additions increase noise linearly Multiplications amplify noise quadratically or worse There is always a precision loss, even without any operations, due to encoding and rescaling features of CKKS	Rescaling (Modulus Switching): After each multiplication, CKKS scales down ciphertext modulus to reduce noise growth and maintain precision—analogue to modulus switching in BGV/BFV Scale Management (ELASM): Dynamically adjusts scales to optimize both noise and latency, guided by SNR metrics and noise waterlines per operation USENIX Bootstrapping (Optional): Resets noise levels when executing deep circuit evaluations, though it's computationally expensive and typically avoided in leveled CKKS
TFHE	Addition and Multiplication	Ring-LWE	IND-CPA	1. NIST has not defined TFHE key sizes like “2048 bits” or “4096 bits” because TFHE relies on lattice parameters, not discrete key lengths 2. The standard TFHE library uses parameters designed for ~ 128 -bit security (LWE $n \approx 630$, ring-LWE $n \approx 1024$) 3. To reach 192- or 256-bit security, you must manually configure higher parameters based on lattice hardness estimators	noise growth is minimal and tightly controlled, unlike leveled HE schemes like BFV, BGV, or CKKS	Gate-by-Gate Bootstrapping: Each logic gate is evaluated using bootstrapping, which resets the error accumulation after every operation FFT-Bias Compensation: Updated bootstrapping techniques account for FFT-induced errors and reduce noise overhead during evaluation Improved Bootstrapping Algorithms: The scheme introduces later bootstrapping procedures that produce smaller evaluation keys and reduced noise growth, compared to legacy methods like AP and GINX while preserving security and efficiency

including data leakage, theft, and performance issues. Cryptographic methods have been widely proposed to secure cloud data. However, despite these advancements, security challenges still hinder full cloud adoption. Many cryptographic systems are resource-intensive and exhibit linear execution times, making them predictable. This has led to a focused review of homomorphic algorithms from 2017 to 2024, examining their types, execution complexity, and energy usage.

Literature review

Cloud security remains a top priority in today's digital landscape. SECtember is a pivotal annual event organized by the Cloud Security Alliance (CSA), a leading authority in cloud and cybersecurity. The conference is designed to improve collaboration and information exchange across the industry. It promotes secure cloud adoption while driving innovation and strategic dialogue. Attendees benefit from expert insights, valuable networking, and tools for navigating complex security challenges. As such, SECtember is a must-attend event for anyone invested in cloud security.

Cloud data storage can be protected using conventional encryption methods like AES, DES, TDES, and RSA. These methods, however, restrict operations on encrypted data unless private keys are exposed to the cloud server. To address this, homomorphic encryption offers a solution by allowing computations on encrypted data without decryption. The decrypted output of such operations matches the result of plaintext operations [105]. It is an asymmetric cryptographic approach used to enhance cloud security. Variants such as Partially Homomorphic Encryption (PHE) and Fully Homomorphic Encryption (FHE) provide different functionalities suited to specific use cases [51, 94].

Based on the cryptographic primitive of homomorphic (partially homomorphic) re-encryption, the study by Ku et al. [54] suggested a privacy-preserving federated learning scheme that can train user data using batch gradient descent (BGD) and protect user data using homomorphic re-encryption. The fog node is used to gather user data, the IoT device encrypts and uploads user data, and the server performs data aggregation and re-encryption. Additionally, the experimental findings and security analysis demonstrate that the approach can finish model training while maintaining local models and user data.

The study by Ku et al. [54] introduced a privacy-preserving federated learning scheme using partially homomorphic re-encryption. It employs batch gradient descent (BGD) for training while protecting user data through encryption and re-encryption. Fog nodes collect the data, IoT devices handle encryption and upload, and the server performs aggregation. The process maintains data confidentiality throughout model training.

Experimental and security evaluations confirm that the scheme preserves both local models and user privacy.

In order to accomplish privacy-preserving and non-interactive gradient descent regression training in machine learning, Wang et al. [99], presented the Privacy-Preserving Multi-party Logistic Regression (PPMLR) technique. The Distributed two Trapdoors Public-Key Cryptosystem (DT-PKC), which fulfills additive homomorphic encryption, is a key component of PPMLR. In order to ensure the secrecy and privacy of the users' private information throughout the training process, the service provider interacts with the cloud server to train the global logistic regression model after the users have gone offline after encrypting their local private data. The paper demonstrates that PPMLR ensures data and model privacy through comprehensive security proof. Two well-known medical datasets from the UCI machine learning library were used in their research. The outcomes of the experiment demonstrate that PPMLR is capable of carrying out privacy-preserving training effectively. The training time of the model is about four times shorter when compared to the state-of-the-art Privacy-Preserving Logistic Regression Algorithm (PPLRA).

Wang et al. [99] introduced the PPMLR technique for privacy-preserving, non-interactive logistic regression training using gradient descent. It leverages the Distributed Two Trapdoors Public-Key Cryptosystem (DT-PKC), which enables additive homomorphic encryption. Users encrypt local data before going offline, while the service provider trains the global model securely with the cloud server. Security analysis confirms PPMLR protects both data and model privacy. Experimental results on UCI medical datasets show that PPMLR outperforms PPLRA, reducing training time by nearly fourfold.

A. Baker [20], suggests a method that aims to improve data protection from several large data sources in order to increase efficiency in providing advanced data protection and, consequently, close security gaps. A partial homomorphic encryption system protects data from hackers by encrypting data produced by several users or sources and processing it in the cloud without decrypting it. Another use for Extremely Distributed Clustering (EDC) is the division of big datasets into several subgroups of cloud computing nodes. This approach can improve the efficacy and performance of big data analytics while simultaneously protecting data and guaranteeing privacy. The findings showed that the suggested method was quicker and provided an improvement in encryption performance of about 23–28%.

A. Baker [20] introduced a method that improves data protection for large, multi-source datasets, targeting increased security and efficiency. The technique uses partial homomorphic encryption to process encrypted data in the cloud without decryption, preventing

unauthorized access. Extremely Distributed Clustering (EDC) further enhances performance by distributing big data across cloud nodes. This combination ensures both data privacy and improved analytics performance. Results show a 23–28% increase in encryption efficiency.

AI-integrated medical technology has improved illness prediction and diagnosis, but data security concerns limit data sharing across organizations. Rao [82] proposed a secure method using a Hybrid Fuzzy Neural Network with Modified Deer Hunt Optimization and Paillier Homomorphic Encryption (FNN-HMDH-PHE). The framework processes datasets through encryption using a unique private key and a fitness function to convert plaintext into ciphertext. CloudSim is used to simulate storage and encryption operations. Experimental results demonstrate the method's superior performance in throughput, efficiency, and encryption/decryption time compared to existing techniques.

Cloud outsourcing offers cost and flexibility benefits, but protecting confidential medical data requires secure encryption. Q. Liu et al., [61] introduced DNA Homomorphic Encryption (DNA-HE) for encrypting medical images in the cloud. Data is first mapped to DNA sequences and then encrypted homomorphically using a key generated by the Rider Optimization Technique. This double encryption ensures security and allows the same process for decryption. The method demonstrated superior performance, reducing encryption time by up to 35% compared to OHE, HE, and ECC algorithms.

Dawson et al. [24] developed the Enhanced Homomorphism Scheme (EHS) based on GPN, LCG, FSWA, and Gentry's homomorphism. The scheme was evaluated on Kaggle data across various parameters, including execution time and homomorphic properties. EHS showed the fastest encryption time at 24 KB and the longest decryption time at 8 KB. Comparatively, EHS outperformed EHES and NFHS in decryption speed at 32 KB, showing a 56.73% improvement. Results suggest that encryption key size influences performance more than data volume.

Mohammed & Abed [66] introduced a fully homomorphic encryption scheme to secure cloud data at rest. The system's security is founded on the difficulty of factoring very large prime numbers using prime modular operations. The secret key is generated using modular arithmetic and Euler coefficient computations. The model increases noise complexity proportionally to the number of users served by the Cloud Service Provider. NIST statistical tests confirmed the algorithm's high randomness and statistical performance.

Min et al. [64] highlighted that while cloud computing improves quality of life, it also introduces privacy and bandwidth concerns. Homomorphic Encryption (HE) offers a promising solution by enabling operations on encrypted data without decryption. However, FHE and

PHE often suffer from low efficiency. To address this, the authors proposed a BGN-type parallel HE method based on the Boneh, Goh, and Nissim cryptosystem, secured by the Learning with Errors (LWE) problem. The method demonstrated up to $5.3\times$ speedup on a 16-core, 4-node cluster, supporting its feasibility in real-world cloud settings.

Using an Algebra Homomorphic Encryption method based on the updated Elgamal (AHEE) algorithm at the client side, Fahina et al. [30] concentrated on offering security to the client for any data saved in the cloud. The digital signature standard DSS, as introduced by NIST in America, has been changed to become AHEE. By using a random value of k in $Eg()$ instead of the same key for encryption, the algorithm's additive homomorphism renders AHEE resistant against plaintext attacks. The subset of the completely homomorphism is the AHEE. It may be used for mobile ciphers, electronic voting, and secure multi-party computing.

Recent homomorphic encryption (HE) developments target secure, efficient data processing in cloud-based Internet of Medical Things (IoMT) systems. Aissaoua et al. [3] proposed an architecture combining blockchain and ultralight linear HE to enable homomorphic operations, particularly addition, over encrypted data. This scheme ensures data confidentiality, traceability, and low energy and communication costs ideal for constrained IoT devices in healthcare settings.

Hangan et al. [40] introduced a privacy-preserving anomaly detection method using the TFHE scheme. By adapting histogram-based detection to work with encrypted IoT data, their system effectively identifies anomalies without requiring decryption. It achieves accuracy comparable to plaintext methods while addressing challenges like input size limits and computational overhead. The study highlights how HE can protect sensitive medical data in cloud environments, supporting secure analytics and decision-making without exposing private information an essential feature for IoMT security and regulatory compliance.

Homomorphic encryption Libraries

Recent literature highlights several advanced homomorphic encryption (HE) libraries supporting secure cloud and IoT applications. Modern homomorphic encryption (HE) libraries have evolved significantly, offering practical solutions for secure data processing across various domains, including cloud computing, healthcare, and database systems.

OpenFHE: Developed collaboratively by institutions like the New Jersey Institute of Technology and Duality Technologies, OpenFHE is an open-source library that supports multiple HE schemes, including BGV and CKKS. It emphasizes modularity and performance,

making it suitable for a range of applications from cloud services to secure data analytics [31].

Hermes: Built atop OpenFHE, Hermes is designed to integrate fully homomorphic encryption into standard SQL engines. It introduces a novel data model that packs multiple records per ciphertext and supports in-place updates and aggregation. This design enables efficient encrypted query processing, achieving significant throughput gains and speedups in data insertion compared to traditional methods [109].

Microsoft SEAL: SEAL is a cross-platform library developed by Microsoft Research, supporting the BFV and CKKS schemes. It offers a user-friendly API and is optimized for performance, making it accessible for developers aiming to implement HE in various applications.

HElib: An open-source library developed by IBM, HElib implements the BGV scheme and includes optimizations like Smart-Vercauteren ciphertext packing techniques. It's designed for efficiency and supports complex operations on encrypted data.

HEAAN: Developed by the Cryptography Lab at Seoul National University, HEAAN focuses on approximate arithmetic operations, making it suitable for applications like machine learning where exact precision isn't always necessary. It implements the CKKS scheme and supports bootstrapping for extended computations.

Evaluating performance of HE in cloud environments

Evaluating the performance of homomorphic encryption (HE) in cloud environments requires attention to core mechanisms like noise growth, modulus switching, relinearization, and bootstrapping. In HE, noise growth accumulates with operations especially multiplication which, if unmanaged, renders ciphertexts indecipherable. Modulus switching mitigates this by transitioning to smaller moduli, though it introduces computational overhead [60].

Relinearization is used after multiplication to reduce ciphertext size, enabling further operations while increasing complexity [52]. Bootstrapping, which resets ciphertext noise to enable continued operations, is one of the most expensive HE operations. Recent innovations such as the Fixed-Point Accelerator for Torus FHE (FPT) have significantly reduced bootstrapping time by using fixed-point arithmetic and streaming architectures [97].

To enhance performance, hardware accelerators like FAB (FPGA-based Accelerator for Bootstrappable FHE) and CiFlow have been introduced, improving bootstrapping and key-switching efficiency through dataflow optimization [72, 107].

Application of Homomorphic Encryption in IoT, IoMT, and cloud infrastructures

Homomorphic Encryption (HE) is a transformative cryptographic technique. Its application is increasingly critical in domains such as the Internet of Things (IoT), Internet of Medical Things (IoMT), and cloud infrastructures, where data privacy, integrity, and security are paramount.

HE in IoMT and Healthcare-Oriented Cloud Systems

In healthcare environments, particularly IoMT, privacy-preserving data analytics is vital. Zhang et al. [108] proposed an efficient FHE-enabled secure cloud-edge computing architecture for IoMT data protection. Their scheme supports secure pandemic modeling by enabling encrypted data processing in the cloud without exposing sensitive patient information.

Similarly, Mantey et al. [63] incorporated HE into a federated learning model for IoMT-based medical recommendation systems. The HE layer ensures that gradient updates remain confidential throughout the learning process while maintaining model performance and accuracy. Hosseingholizadeh et al. [43] introduced a homomorphic multi-party computation (HMPC) protocol tailored for IoMT environments. Their approach enables secure aggregation and processing of encrypted data from multiple sources without requiring interactive decryption or shared keys, thus enhancing both security and scalability.

HE in IoT and Cloud Infrastructure

Lakhan and Wu [58] presented a blockchain-based homomorphic encryption framework for secure intelligent transport systems in fog/cloud-integrated IoT networks. The solution allows encrypted transport data to be verified and processed securely across distributed nodes, ensuring privacy, authenticity, and real-time operability. In broader IoT-cloud settings, the integration of homomorphic encryption facilitates secure data analytics and search mechanisms. Multi-key HE has been applied to allow secure searches over encrypted data owned by multiple entities without central key sharing, thus improving privacy in decentralized IoT environments [43].

Benefits and challenges

Benefits:

- **Confidential Data Processing:** HE ensures that sensitive IoT and healthcare data remain encrypted during processing, preserving privacy end-to-end.
- **Regulatory Compliance:** HE aids compliance with regulations like HIPAA and GDPR by preventing unauthorized access to patient or personal data.

- Secure Collaboration: Multi-owner and federated HE models enable collaborative analytics without compromising individual data privacy.

Challenges:

- High Computational Cost: Fully Homomorphic Encryption schemes are computationally intensive and not yet optimal for resource-constrained IoT devices.
- Latency and Efficiency: Real-time HE processing is difficult, especially for latency-sensitive applications like autonomous transport systems.
- Complex Key Management: Multi-key encryption schemes require advanced key management and synchronization to operate securely at scale.

Related research

Cloud computing offers advantages such as cost savings, resource pooling, and simplified administration, driving its widespread adoption. However, transferring client data and applications to shared cloud environments introduces security risks like data theft and exposure of personal information. To mitigate these threats, encryption has become a key strategy. Alemami et al. [5] evaluated encryption algorithms IDEA, Blowfish, RSA, DES, and AES based on factors like security, memory usage, and encryption time. Their study concluded that AES, Blowfish, and DES are more secure than RSA and IDEA, with Blowfish being memory-efficient and AES excelling in speed and large-scale data encryption.

Kwao Dawson et al. [57] reported that the increasing volume of Industrial data has led to the adoption of cloud computing, although it introduces both opportunities and challenges. Despite various cryptographic algorithms being proposed, data security remains a key concern due to high CPU usage and bandwidth demands during encryption of large datasets. The authors reviewed 73 papers from 2016 to 2022, selected from major academic databases using PRISMA guidelines. Their survey found that encryption is the most widely used solution for data and cloud security. It also showed that 90% of security methods follow linear runtime patterns, while nonlinear symmetric stream encryption remains underutilized.

Alonso et al. [7] highlighted that the growing demand for compute and storage, driven by cloud computing and IoT, makes traditional single-provider cloud models inadequate due to risks like vendor lock-in. Their study advocates for a multi-cloud approach, influencing application development, deployment, and operation across heterogeneous platforms. They aimed to define multi-cloud native applications, architecturally characterize them, and identify unresolved research questions through a systematic literature review of studies from 2011 to

2021. The review revealed five main research themes to improve the DevOps lifecycle of multi-cloud-native applications. The study concludes with recommendations for further research in this evolving area.

Abrera [1], conducted a comprehensive analysis of cloud computing security and data privacy concerns by reviewing academic papers, industry reports, and conference proceedings. The study examined areas such as authentication, access control, encryption, leak prevention, and compliance. Key challenges identified include data breaches, key management, unauthorized access, opaque provider policies, and regulatory difficulties. Solutions proposed include intrusion detection systems, encryption techniques, access control models, and privacy-preserving data mining. The study also suggests future research should focus on advanced technologies like blockchain and homomorphic encryption to further enhance cloud data security.

Gong et al. [37] reviewed recent advances in Fully Homomorphic Encryption (FHE) acceleration techniques to address its high computational and storage costs. The review analyzes studies published between 2019 and 2022. It provides a detailed overview of the latest findings and classifies acceleration strategies into hardware-based and algorithmic categories. The authors also compare various methods and propose evaluation metrics for assessing performance. Their findings highlight current challenges and opportunities in optimizing FHE. The paper concludes with future research directions to guide further development and application of FHE acceleration.

Hussain et al. [45] discuss how the rise of Internet usage has increased both the number of users and data, leading to higher demands for flexible computing access. Traditional infrastructures cannot efficiently meet these demands due to their inflexibility and cost. Cloud computing addresses this by enabling on-demand access to computing services, but it introduces new privacy and security risks. The study presents a systematic model of cloud computing and identifies key threats, such as unauthorized access, data leaks, and privacy violations. It evaluates storage privacy and security concerns and explores advanced cryptographic methods, including ABE, HE, and SE. The paper concludes by outlining unresolved security issues and suggesting future research directions to strengthen cloud computing protections.

Munjal & Bhatia, [68], discuss how cloud computing and storage have transformed data processing by improving resource accessibility and reducing labor. Customers outsource data processing by transferring data to the cloud, where third-party providers gain access to private information. In healthcare, patient records are stored and processed in data centers, but traditional encryption requires data decryption before use, risking sensitive information exposure. Homomorphic encryption

addresses this by allowing computations on encrypted data, keeping it secure from service providers. The study reviews the classification and evolution of homomorphic cryptosystems. It also highlights their significant contributions to securing healthcare data.

Yang et al. [106] highlight how biometric technology is widely used in law enforcement, healthcare, and financial verification. Due to the permanence and uniqueness of biometric traits, securing biometric data is critical. Security and privacy are essential for ensuring integrity and availability in biometric applications. Homomorphic encryption (HE) is explored as a solution to protect biometric data within the cryptographic domain. The study thoroughly reviews various HE methods tailored to different biometric features. It also discusses integrating HE with emerging technologies like blockchain and machine/deep learning, while identifying current challenges and future research directions.

Janet Julia Ang'udi, [14] explores key cloud security topics including network security, access control, data breaches, and compliance with legal frameworks. The study analyzes real-world case studies to illustrate the impact of cloud security breaches and how to address them. It offers insights into the complexities of cloud security risks and mitigation strategies. The research highlights the potential of AI and machine learning to enhance cloud security through proactive threat detection and automated responses. It also provides a forward-looking view on emerging trends and challenges in cloud security. The authors aim to guide organizations in adopting effective strategies and best practices to balance cloud computing benefits with strong security protections.

According to Fatima & Ahmad [33] Cloud computing refers to highly flexible processing. Applications, collections, and platforms to support governments, businesses, and individuals. In light of this, SMB (Small and Medium Business) organizations are constantly modifying cloud processing services in an effort to reduce costs and improve business efficiency. Although the benefits and energy of cloud administration are clear, cloud computing security is increasingly a greater concern. "What amount secure is cloud computing condition?" observed that one of the main barriers to the further development of cloud computing is security. People are hesitant to transmit their data and apps under cloud conditions due to a number of serious security risks and problems. The article's main objective is to identify the importance of security risks and concerns that must be taken into account while transmitting and developing cloud-based services, as well as the best way to mitigate those risks and concerns. However, it's important to understand that cloud computing isn't inherently unstable, it only has to be monitored and accessed securely.

Fatima & Ahmad [33], describe cloud computing as a flexible platform supporting governments, businesses, and individuals. SMBs continually adapt cloud services to reduce costs and enhance efficiency. Despite its benefits, security remains a major concern limiting cloud adoption. The authors highlight that uncertainty around cloud security discourages users from trusting cloud environments with their data and applications. The article focuses on identifying critical security risks and proposing mitigation strategies for cloud services. It concludes by emphasizing that cloud computing itself is not insecure but requires secure monitoring and access controls.

Bootstrapping was originally introduced by Craig Gentry in his groundbreaking 2009 paper as a method to refresh ciphertexts by homomorphically evaluating the decryption circuit of the encryption scheme itself. This innovation enabled the first construction of a fully homomorphic encryption (FHE) scheme, allowing for arbitrary computations on encrypted data without decryption.

Advantages of Bootstrapping.

- Enables Full Homomorphism: Makes it possible to evaluate arbitrarily deep circuits by reducing accumulated noise.
- Security Assurance: Helps preserve ciphertext integrity over extended computations.
- Foundation for FHE Schemes: A critical step toward achieving practical FHE.

Disadvantages of Bootstrapping.

- High Computational Overhead: Bootstrapping is typically very slow and resource-intensive.
- Complex Implementation: Requires intricate handling of encrypted decryption circuits.
- Not Always Necessary: Some leveled HE schemes avoid bootstrapping by limiting computation depth.

Methodology

This study considers the quantity of works on cloud data privacy protection with homomorphic algorithm, interprets them using publications on the topic, and creates a systematic literature review using PRISMA.

Research questions

This work aims to critically examine homomorphic algorithms and their application in safeguarding cloud data privacy. Security interventions by researchers are also considered. Six goals are taken into consideration in this study.

1. What homomorphic algorithm was used to protect the privacy of cloud data in the article?

2. How did the algorithm do the encryption and decryption?
3. Did the article measure the nonlinear and linear time execution of the homomorphic algorithm?
4. Measure of the communication overhead of the algorithm?
5. What is the energy consumption and resource optimization of the algorithm?
6. What is the negative effect of the complexity of the encryption scheme on the algorithm's performance.

Approaches for accessing articles

In order to address the research issues, this part concentrates on the various terminology, databases, reference materials, and search techniques. As follows, this has been thoroughly addressed.

Phrases used

Keyword searching using the phrase "Cloud data privacy protection with homomorphic algorithms" was used across multiple databases to find the various publications used in this study.

Electronic sources

To identify papers for this study, a number of popular online research databases were used, including IEEE Xplore, Google Scholar, Scopus, and Taylor & Francis.

Reference management

A significant number of articles were downloaded based on the search criteria. The referencing management technology used was the Mendeley reference generation tool.

Search processes

The well-known databases were searched for literature on the subject. This included conference proceedings, books, and journal articles. After downloading all 207 papers, they were arranged in Mendeley for convenience of reading and referencing. The data are then shown in a PRISMA framework in Figs. 8 and 9 [81].

75 out of 207 papers were judged pertinent to the topic at hand after the publications were categorized using a selection procedure based on their center of interest. The following exclusion process was employed to choose the papers of interest:

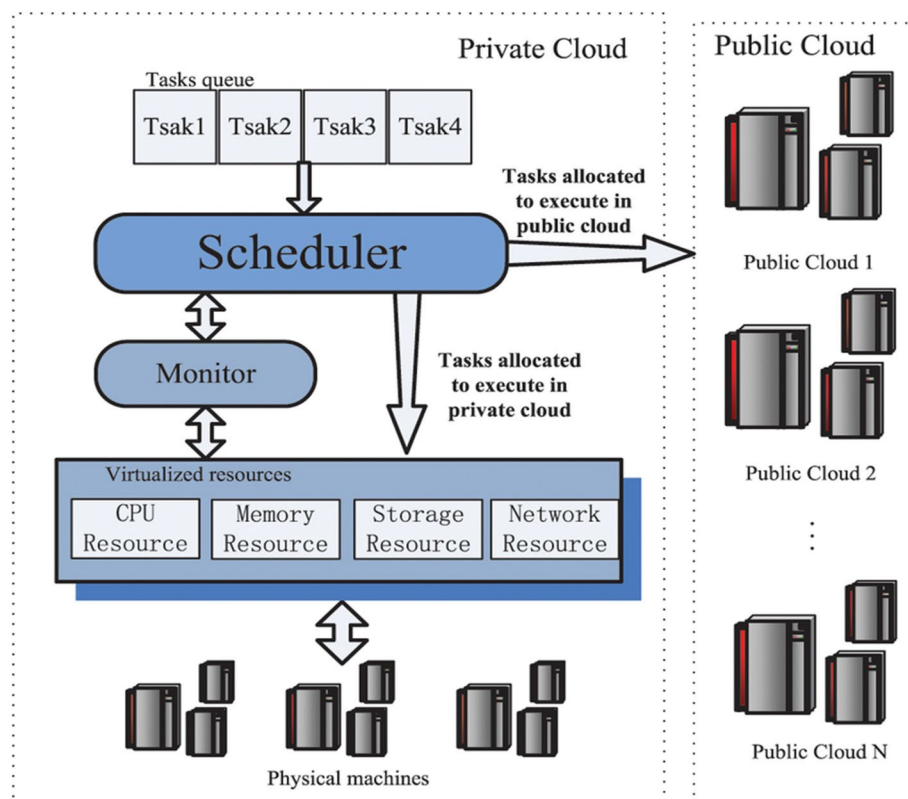


Fig. 8 Task scheduling in a hybrid cloud environment [110]

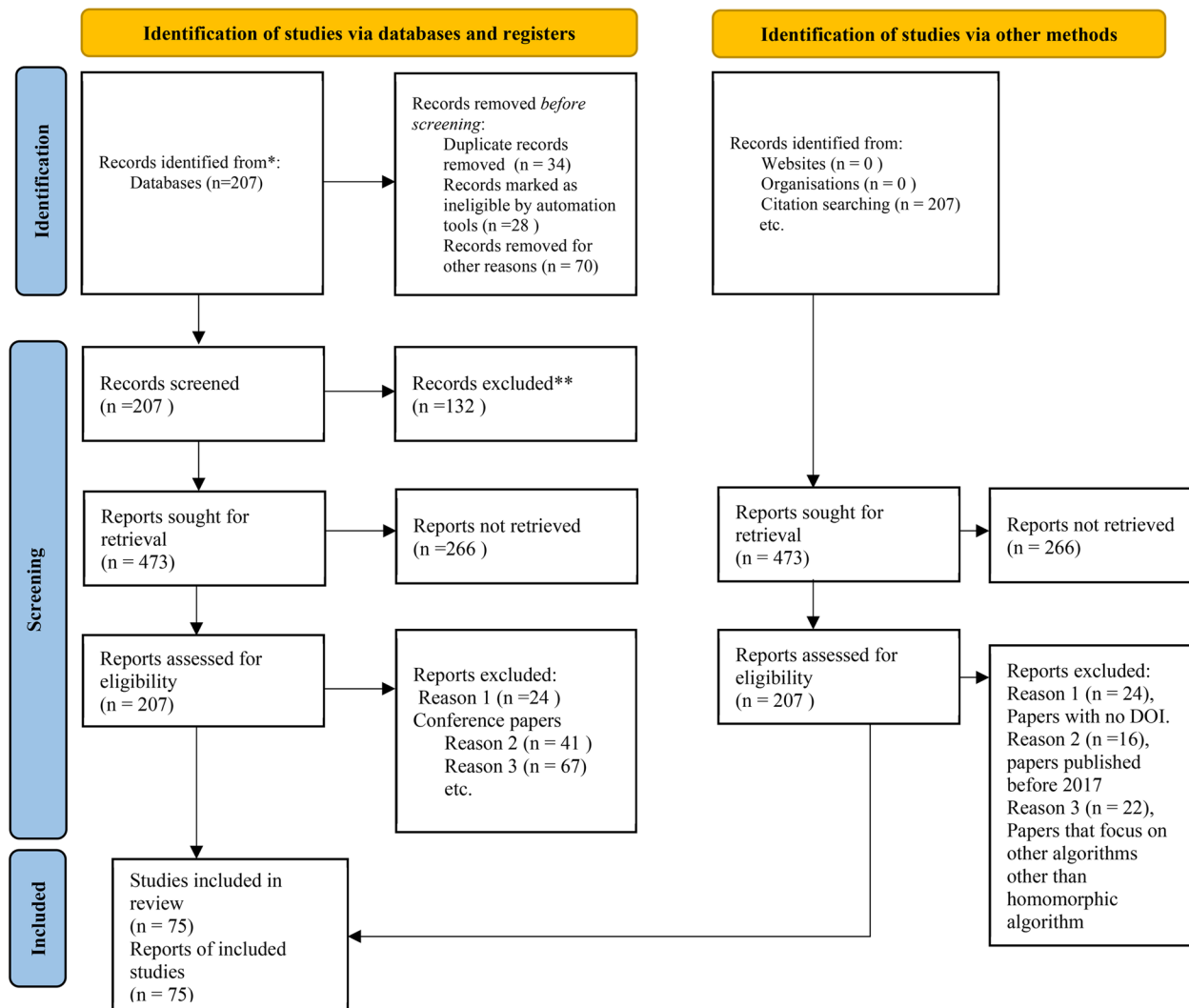


Fig. 9 Flow diagram for the systematic review which included searches of databases and other sources [81]

Table 5 Homomorphic algorithm and the number of articles

Type of Homomorphic Algorithm	Article	Summary
Fully Homomorphic (FHE)	P8,P10,P11,P13,P15,P16,P18,P20,P27,P28,P32,P35,P38,P39,P40,P41,P43,P44,P45,P46,P47,P48,P49,P50,P51,P52,P54,P55,P56,P57,P58,P59,P62,P63,P64,P67,P71,P73	38
Partially Homomorphic (PHE)	P1,P2,P3,P4,P5,P6,P7,P21,P22,P24,P30,P33,P34,P42,P51,P60,P65,P66,P68,P69,P70,P72,P74,P75	24
Somewhat Homomorphic (SWHE)	P9,P17,P29,P37	4
Not specifically stated (NSS)	P12,P14,P19,P23,P25,P26,P31,P36,P53,P54,P61	11

- (I) Conference papers.
- (II) Papers published before 2017,
- (III) Papers that focus on other algorithms other than homomorphic algorithm.
- (IV) papers without a DOI.
- (V) Papers with anonymous citations.
- (VI) Papers that focus on technologies other than the data privacy protection of cloud computing.
- (VII) Papers focusing on cloud taxonomy.

Results

The results of the systematic literature review are presented in this part, and their commentary is provided in the supplemental subsections. Table 5 categorizes the list of publications, full homomorphic, partial homomorphic,

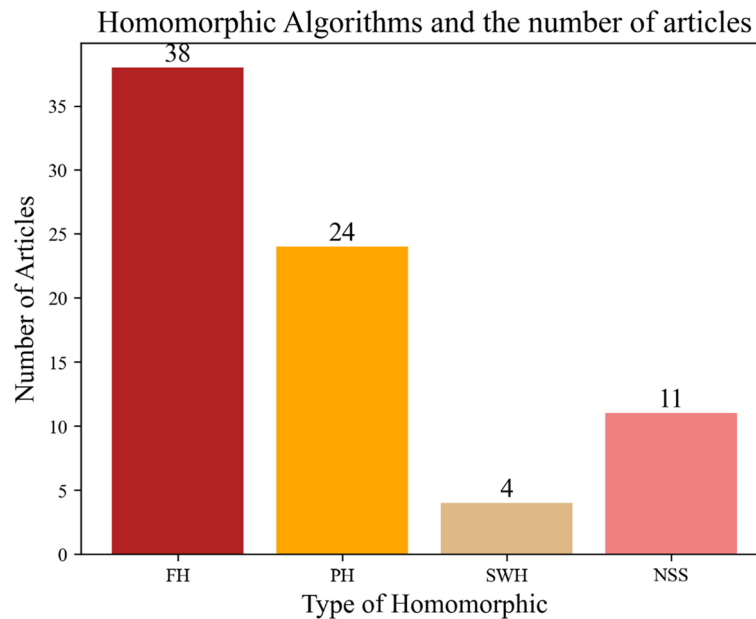


Fig. 10 Shows the number of articles that made use of each of the type of homomorphic algorithm

Table 6 Partially Homomorphic encryption scheme

		Summary
Not categorical or other variants	P1,P68,P70	3
Additive	P2,P3,P4,P5,P6,P7,P21,P22,P24,P30,P33,P34,P42,P60,P65,P66,P69,P72,P74,	19
Multiplicative	P51,P75	2

and the run time trend of these homomorphic algorithms. The table provides clear guidelines for academics looking for answers to cloud data protection with homomorphic algorithms. There is also the concern that, despite the benefits of this algorithm, its computational overhead and complexity is making it difficult for it implementation in real time applications.

What homomorphic algorithm was used to protect the privacy of cloud data in the article?

The data extraction table, which is attached to this article as a supplementary document shows a number of proposed variants of homomorphic algorithm. According to Fig. 10 and Table 5, the adoption of Fully homomorphic algorithm, which accounts for 38 number of publications from 2017 to 2024, is the most popular for ensuring cloud data privacy protection. It is was followed by the usage of Partially Homomorphic algorithm, which represented 24 articles of the total 75. 11 articles did not specifically state the type of homomorphic algorithm use to protect the cloud data even though it categorically mentioned the use of Homomorphic algorithm. Somewhat Homomorphic, 4 article made use of this type of algorithm.

PHE Scheme and percentage of article

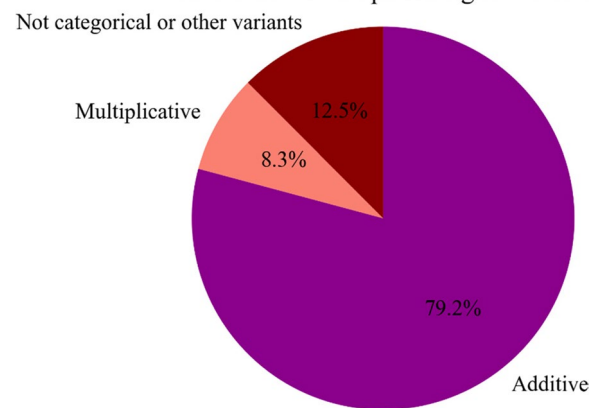


Fig. 11 Partially Homomorphic encryption techniques and percentage of articles that used them

How did the algorithm do the encryption and decryption?

Fully Homomorphic Encryption (FHE) supports both additive and multiplicative operations on encrypted data, while Partially Homomorphic Encryption (PHE) supports only one. The Paillier and RSA schemes are examples of additive and multiplicative PHE, respectively. Out of 24 articles using PHE, 79.2% focused on additive PHE, 8.3% on multiplicative PHE, and 12.5% did not specify the variant. Multi-key Homomorphic Encryption (MKHE) also appears, allowing operations on ciphertexts encrypted with different keys. These distinctions clarify the strengths and limitations of homomorphic encryption methods. This understanding is essential for applying these schemes in cloud computing and privacy-preserving data processing (Table 6) Fig. 11.

Did the article measure the nonlinear time execution of the homomorphic algorithm?

The execution time of an algorithm is the time it takes to complete its process from start to finish. It is a critical measure of an algorithm's efficiency, typically calculated as time complexity in theoretical analysis or as wall-clock time in practical implementations.

Table 7 Time execution

Linear		Non-linear		Not Categorical
Measured	Not Measured	Measured	Not Measured	
P2,P4,P5,P7,P9,P11,P12,P13,P14,P15,P16,P18,P19,P20,P22,P23,P24,P25,P27,P28,P33,P35,P37,P38,P39,P40,P41,P42,P43,P44,P45,P49,P50,P53,P54,P55,P56,P57,P60,P62,P63,P65,P67,P68,P69,P71,P73	P1,P3,P17,P2,P25,P29,P30,P31,P32,P34,P36,P39,P46,P47,P48,P51,P52,P58,P59,P61,P64,P70,P72,P74,P75	P3,P4,P7,19,P24,P25,P38,P50	P1,P2,P5,P6,P9,P11,P13,P14,P15,P16,P17,P18,P20,P21,P22,P26,P27,P28,P29,P30,P31,P32,P34,P35,P36,P39,P40,P41,P43,P44,P45,P47,P48,P49,P52,P53,P54,P56,P57,P58,P59,P60,P61,P62,P64,P65,P66,P68,P69,P70,P71,P72,P73,P74,P75	P1,P6,P8,P10
48	26	7	63	4

Linear time execution of an algorithm means that the time it takes to complete its work grows proportionally with the size of the input, denoted as (n) in Big O notation. In other words, if the input size doubles, the execution time of a linear-time algorithm will also roughly double. Table 7 summarizes the execution times of the algorithms proposed by each selected article. Forty-eight articles measured linear execution, representing 64.9% of the total 75 articles, highlighting the importance of measuring linear-time execution. Thirty-five point one percent, or 26 articles, did not measure the linear execution time of the proposed algorithms.

Nonlinear time execution

Nonlinear time execution of an algorithm refers to algorithms whose execution time grows faster than linearly with respect to the size of the input. Common types of nonlinear time complexities include quadratic $O(n^2)$, cubic $O(n^3)$ and log-linear $O(n \log n)$ complexities, among others. From Fig. 12 and Table 7, approximately 10% of the total 75 articles measured the nonlinear performance of their proposed algorithms while 90% did not carryout any measurement or investigation on the nonlinear performance of the proposed algorithm.

Measure of the communication overhead of the algorithm?

The communication overhead of an algorithm refers to the additional time and resources required for data exchange between components in a system, which can significantly impact performance. This overhead is influenced by factors such as the number of messages exchanged, the size of the messages, and the communication patterns employed. Understanding and minimizing communication overhead is crucial for optimizing algorithm efficiency, especially in distributed and parallel computing environments [101] (Fig. 13).

The first column of Table 8 show the individual articles that measured the computational and communication overhead of their proposed algorithms. In terms of percentage Fig. 14 shows that 44% representing 33 articles out of the 75 measured the computational and communication overhead, while 56% representing 42 articles did not (Figs. 15, 16, 17).

What is the energy consumption and resource optimization of the algorithm?

Energy consumption and resource optimization in cloud computing are critical for enhancing efficiency and sustainability. Suba Santhosi G B, et al. (2023) proposed an algorithm that reduces energy consumption by 30% compared to load balancing round robin and ant colony algorithms, while optimizing resource allocation and

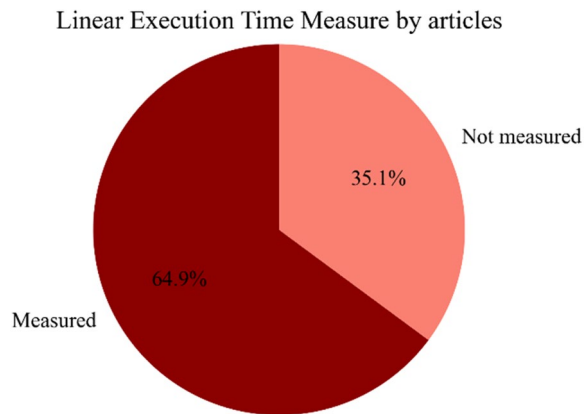


Fig. 12 Linear Time Execution of the proposed algorithm of each article

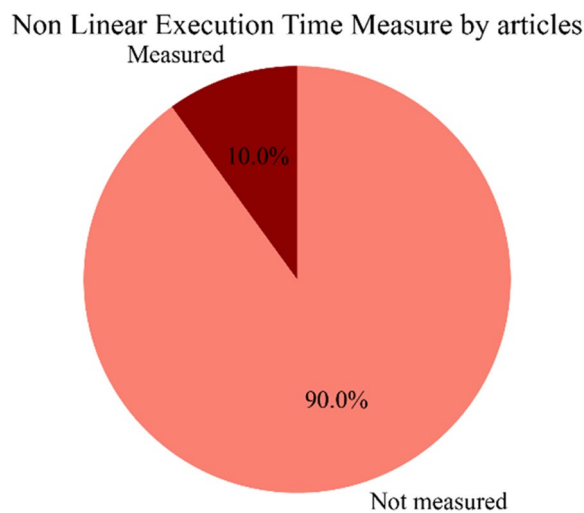


Fig. 13 Nonlinear time execution of the proposed algorithms in the articles expressed in percentage

improving task response time. From Table 8, only one article measured the energy consumption of its proposed algorithm, with the remaining 74 not measuring it. This shows a knowledge gap that needs to be addressed regarding the energy consumption of cloud data protection algorithms.

What is the negative effect of the complexity of the encryption scheme on the algorithm's performance

The complexity of an encryption scheme significantly impacts an algorithm's performance, influencing both security and computational efficiency. As encryption complexity increases, processing time and resource consumption typically rise, leading to potential trade-offs between security and performance. While complex encryption schemes provide enhanced security, they can also cause inefficiencies that may not be suitable for all applications. Simpler algorithms may offer adequate security with better performance, highlighting the need for a balanced approach in selecting encryption schemes.

Table 8, showed that 14 articles representing 18.7% of the 75 articles measured how the complexity of encryption affects their algorithms performance. Fifteen articles representing 20% according to Fig. 10 discussed the complexity of encryption without measuring how it impacts the algorithms performance while 61.3%, thus 46 articles did not measure the effect of complexity of encryption on their proposed algorithm performances which shows a gap in the research.

Importance of evaluation metrics

The evaluation metrics used to assess algorithm performance are crucial for ensuring accurate and reliable outcomes across various applications. Different metrics can yield significantly different rankings of algorithms, underscoring the need for a comprehensive evaluation approach. Common evaluation metrics used to measure algorithm performance include accuracy, execution time, computational overhead, communication overhead, security analysis, and recognition efficacy (Table 9).

Over the last decade, organizations and institutions have moved their data storage to the cloud, resulting in significant cost savings. Though this shift offers many benefits, it has also raised concerns about security. Homomorphic algorithms have gained widespread adoption due to their ability to allow computations on encrypted data without decryption. Table 7 presents the number of selected articles per year, including one article of interest with an unknown publication year. Figure 11 is a bar chart illustrating the publication years alongside the corresponding number of selected articles. The years 2022 and 2023 had the highest number of articles meeting the inclusion criteria, with 21 each, followed closely by 2024 with 18 articles indicating a growing research interest in using homomorphic algorithms to protect cloud data privacy.

A word cloud is a graphical representation of text data in which the size of each word reflects its frequency or relevance. Words that appear more frequently in the dataset are shown in larger, bolder fonts. For example, Fig. 12 demonstrates that terms such as *cloud*, *data*, and *homomorphic encryption* are highly prominent when analyzing keywords from the selected articles. This visualization makes it easy to quickly identify key concepts. It effectively condenses frequently used terms in textual data for easier interpretation.

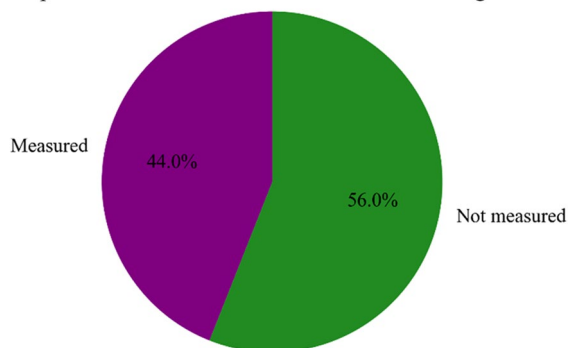
Limitations

The systematic literature review is limited to publications from 2017 to 2024. While the paper emphasizes homomorphic encryption as a solution for data privacy, it may not comprehensively address other encryption methods or security measures that could also be relevant in the context of cloud computing. The findings

Table 8 Showing some important metrics that impact the performance of an algorithm

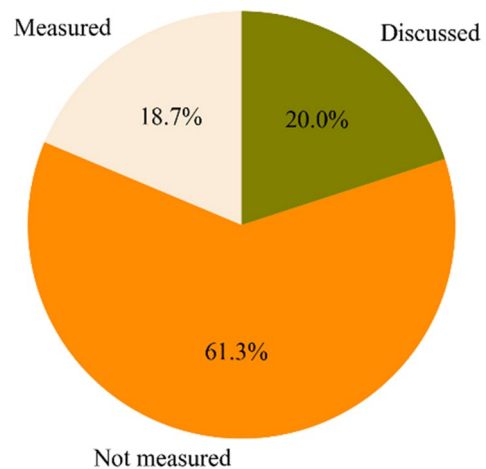
Computational and Communication overhead of algorithm		Energy consumption of algorithm		Effect of complexity of encryption on algorithm		
Measured	Not Measured	Measured	Not Measured	Measured	Not Measured	Discussed
P1,P2,P3,P5,P6,P7,P13,P18,P19,P23,P24,P27,P34,P36,P38,P39,P40,P41,P42,P46,P48,P49,P50,P51,P53,P54,P56,P60,P61,P63,P64,P66,P68	P4,P8,P9,P10,P11,P12,P14,P15,P16,P17,P20,P21,P22,P25,P26,P28,P29,P30,P31,P32,P33,P35,P37,P43,P44,P45,P47,P52,P55,P57,P58,P59,P62,P65,P67,P69,P70,P71,P72,P73,P74,P75	P42	P1,P2,P3,P4,P5,P6,P7,P8,P9,P10,P11,P12,P13,P14,P15,P16,P17,P18,P19,P20,P21,P22,P23,P24,P25,P26,P27,P28,P29,P30,P31,P32,P33,P34,P35,P36,P37,P38,P39,P40,P41,P42,P43,P44,P45,P46,P47,P48,P49,P50,P51,P52,P53,P54,P55,P56,P57,P58,P59,P60,P61,P62,P63,P64,P65,P66,P67,P68,P69,P70,P71,P72,P73,P74,P75	P1,P3,P9,P14,P18,P38,P39,P41,P42,P46,P57,P59,P60,P68	P6,P7,P12,P13,P15,P16,P17,P21,P22,P23,P24,P26,P27,P28,P29,P30,P31,P32,P33,P34,P35,P36,P37,P40,P43,P44,P45,P47,P49,P54,P55,P56,P58,P61,P62,P63,P64,P65,P66,P67,P69,P71,P72,P73,P74,P75	P2,P4,P5,P8,P10,P19,P20,P25,P48,P51,P52,P53,P70
33	42	1	74	14	46	15

Computational and Communication overhead of algorithm

**Fig. 14** Computational and communication overhead measure by articles

and recommendations derived from the literature review may not be universally applicable across all cloud service models or industries, as different sectors have unique security requirements and challenges. The selection of studies for the review may introduce bias, as it relies on keyword searches and the availability of published papers, which might not capture all relevant research or perspectives on cloud data privacy. The paper may not fully account for the practical limitations and challenges of implementing homomorphic encryption in real-world

Effect of complexity of encryption on algorithm

**Fig. 15** Measure of effect of complexity of encryption on algorithm performance

cloud environments, such as computational overhead and performance constraints. Given the rapid evolution of cloud technologies and security threats, the findings may quickly become outdated, necessitating ongoing

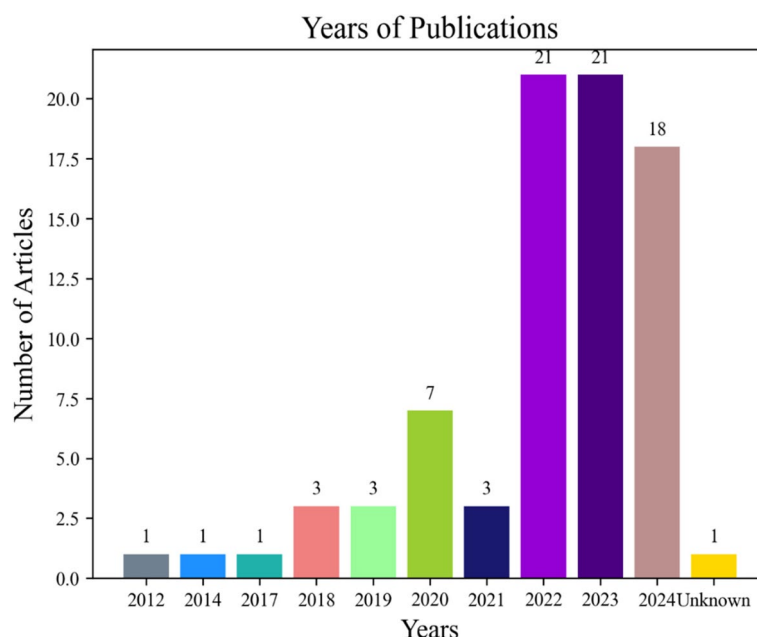


Fig. 16 Years and the number of selected article

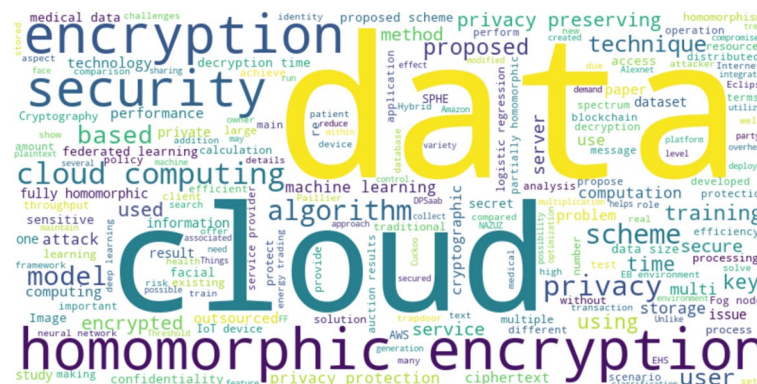


Fig. 17 Keywords word cloud for the selected articles

Table 9 Number of articles and their publication year

2017	2018	2019	2020	2021	2022	2023	2024
2	3	4	9	3	21	21	18

research to keep pace with new developments. These limitations highlight the need for further investigation and advancement in the field of cloud data privacy and encryption techniques to address the evolving landscape of cloud computing.

Conclusion

The systematic review on *Cloud Data Privacy Protection with Homomorphic Algorithm* underscores the critical importance of safeguarding data privacy in the rapidly evolving landscape of cloud computing. As organizations increasingly rely on cloud services for data storage and processing, the associated security and privacy

challenges become paramount. This review meticulously examines the body of literature surrounding homomorphic encryption a promising solution that allows computations to be performed on encrypted data without compromising confidentiality. The findings reveal that Fully Homomorphic Encryption (FHE) has emerged as the most widely discussed approach in the literature, with a significant number of publications highlighting its potential for enhancing data privacy in cloud environments. However, the review also identifies substantial challenges, including the computational overhead and complexity associated with implementing homomorphic algorithms in real-time applications. These limitations

pose significant barriers to the widespread adoption of FHE and other homomorphic techniques, necessitating further research and innovation to develop more efficient and practical solutions.

Moreover, the review emphasizes the need for a multi-faceted approach to cloud data security that encompasses not only encryption but also robust authentication, access control, and compliance with regulatory frameworks. The insights gathered from the literature provide valuable recommendations for researchers, practitioners, and policymakers aiming to enhance data privacy and security protocols in cloud computing settings. This review serves as a foundational resource for understanding the current state of research on cloud data privacy protection through homomorphic algorithms. It calls for collaborative efforts among academia, industry, and regulatory bodies to address the pressing challenges of data security in the cloud, ultimately fostering a safer and more secure digital environment for all stakeholders involved.

Recommendations

1. Future research should focus on optimizing fully and partially homomorphic encryption algorithms to reduce computational overhead and enhance performance. This includes developing new mathematical frameworks, leveraging acceleration techniques, and exploring hybrid models.
2. Researchers should investigate the integration of homomorphic encryption with emerging technologies such as blockchain, artificial intelligence, and machine learning. Such combinations could yield robust, scalable data privacy solutions for cloud environments without compromising confidentiality.
3. The development of energy-efficient homomorphic encryption algorithms should be prioritized by optimizing computational processes to reduce energy consumption and support sustainable cloud computing.
4. Establishing comprehensive security frameworks is essential for strengthening cloud security. These should incorporate homomorphic encryption, access control mechanisms, authentication protocols, and compliance with data protection regulations. Future studies should aim to develop holistic models that provide practical guidelines for implementation.
5. Future research should emphasize real-world implementations of homomorphic encryption across various industries. Conducting empirical case studies will help assess effectiveness, scalability, and user experience in live cloud environments,

thereby refining algorithms for broader adoption and identifying practical challenges and best practices.

By implementing these recommendations, the energy consumption associated with homomorphic algorithms can be effectively managed, contributing to more sustainable and efficient cloud computing practices while maintaining robust data privacy protections.

Abbreviation

HE	Homomorphic Encryption
FHE	Fully Homomorphic Encryption
NIST	Cloud Computing definition that provided by National Institute of Standards and Technology
AWS	Amazon Web Services
GCP	Google Cloud Platform
SaaS	Software as a Service
Paas	Platform as a Service
IaaS	Infrastructure as a Service
CSP	Cloud Service Provider
TEEs	Trusted Execution Environments
IAM	Identity and Access Management
VPCs	Virtual Private Clouds
IDS	Intrusion Detection Systems
MFA	Multi-Factor Authentication

Authors' contributions

MAJ: Conceived the idea of conducting the research and gathered all the primary papers from research databases. PA: Contributed in paper selection and reviewing of articles to meet the selection criteria and playing a supervisory role. OA: Editing of the work, helping in writing of the findings and playing a supervisory role. KA: Editing of the work and playing of supervisory role.

Funding

This research did not receive funding from any source.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Received: 21 March 2025 / Accepted: 15 August 2025

Published online: 22 November 2025

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